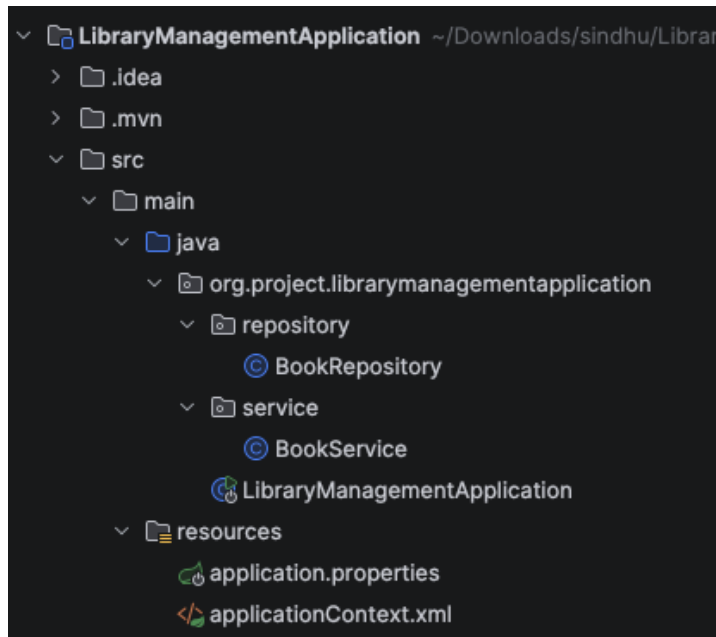


Week2-SPRING CORE & MAVEN:

TASK1:



```
1 package org.project.librarymanagementapplication;
2
3 import ...
4
5
6
7
8
9
10 @SpringBootApplication
11 public class LibraryManagementApplication {
12
13     public static void main(String[] args) {
14         ApplicationContext context= new ClassPathXmlApplicationContext( configLocation: "applicationco
15
16         BookService service =context.getBean( name: "bookService",BookService.class);
17
18         BookRepository repo=context.getBean( name: "bookRepository", BookRepository.class);
19
20         service.displayService();
21         repo.displayRepository();
22
23         //SpringApplication.run(LibraryManagementApplication.class, args);
24     }
25
26
27
```

```
/Users/siddharth/Library/Java/JavaVirtualMachines/ms-21.0.8/Contents/Home/bin
Book Service Created
Book Repository Created

Process finished with exit code 0
```

```
LibraryManagementApplication.java  BookRepository.java  BookService.java
package org.project.librarymanagementapplication.repository;

public class BookRepository {
    public void displayRepository(){
        System.out.println("Book Repository Created");
    }
}
```

```
package org.project.librarymanagementapplication.service;

public class BookService {
    public void displayService(){
        System.out.println("Book Service Created");
    }
}
```

EXERCISE 2:

```
LibraryManagementApplication.java  BookRepository.java  BookService.java  application.properties

1  <?xml version="1.0" encoding="UTF-8"?>
2
3  <beans xmlns="http://www.springframework.org/schema/beans"
4      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5      xsi:schemaLocation="
6          http://www.springframework.org/schema/beans
7          https://www.springframework.org/schema/beans/spring-beans.xsd">
8
9      <bean id="bookRepository"
10         class="org.project.librarymanagementapplication.repository.BookRepository"/>
11
12      <bean id="bookService"
13         class="org.project.librarymanagementapplication.service.BookService">
14
15         <property name="bookRepository"
16             ref="bookRepository"/>
17     </bean>
18
19 </beans>
```

```
Book Service Created
Book Repository : Fetching book details..

Process finished with exit code 0
```

```
package org.project.librarymanagementapplication;

import ...

@SpringBootApplication
public class LibraryManagementApplication {

    public static void main(String[] args) {
        ApplicationContext context= new ClassPathXmlApplicationContext( "applicationcontext.xml");

        BookService service =context.getBean( name: "bookService",BookService.class);

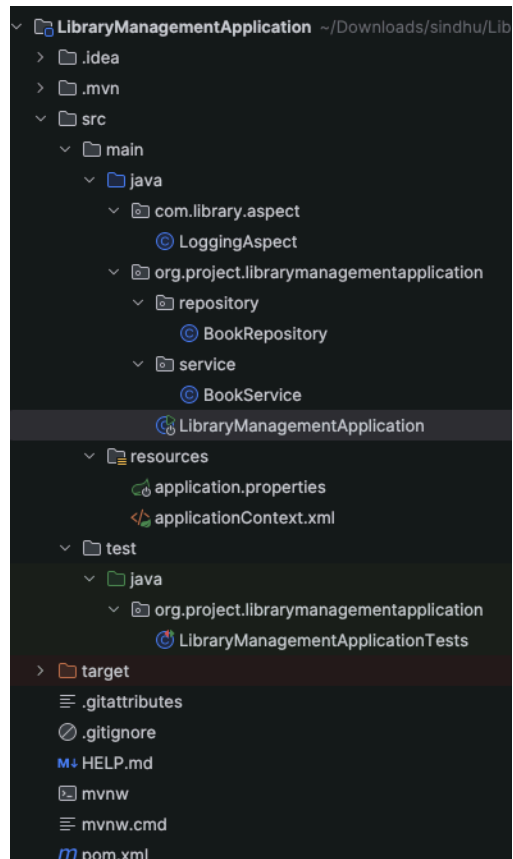
        BookRepository repo=context.getBean( name: "bookRepository", BookRepository.class);

        service.displayService();

        repo.displayRepository();

        //SpringApplication.run(LibraryManagementApplication.class, args);
    }
}
```

EXERCISE:3



```
package org.project.librarymanagementapplication;

import ...

@SpringBootApplication
public class LibraryManagementApplication {

    public static void main(String[] args) {
        ApplicationContext context= new ClassPathXmlApplicationContext( configLocation: "applicationcontext.xml

        BookService service =context.getBean( name: "bookService",BookService.class);

        BookRepository repo=context.getBean( name: "bookRepository", BookRepository.class);

        service.displayService();

        repo.displayRepository();

        //SpringApplication.run(LibraryManagementApplication.class, args);
    }
}
```

```
?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:aop="http://www.springframework.org/schema/aop"
       xsi:schemaLocation="http://www.springframework.org/schema/beans
                           http://www.springframework.org/schema/beans/spring-beans.xsd http://www.springframework.org/schema/aop http://www.springframework.org/schema/aop/spring-aop.xsd" >

    <bean id="bookRepository"
          class="org.project.librarymanagementapplication.repository.BookRepository"/>

    <bean id="bookService"
          class="org.project.librarymanagementapplication.service.BookService">
        <property name="bookRepository"
                  ref="bookRepository"/>
    </bean>

    <bean id="loggingAspect"
          class="com.library.aspect.LoggingAspect"/>

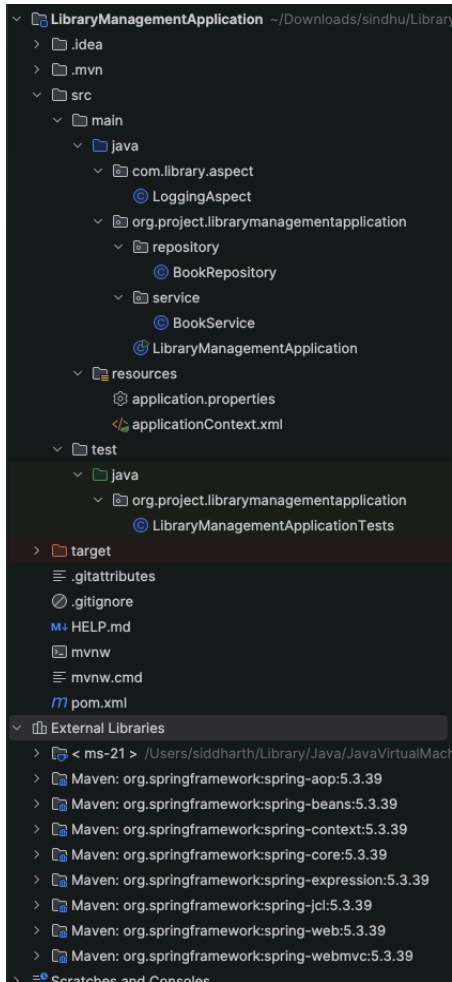
    <!-- Enable AOP -->
    <aop:aspectj-autoproxy/>

```

```
/Users/siddharth/Library/Java/JavaVirtualMachines/ms-21.0.8/Contents/Home/bin/java ...
Before executing : displayService
Book Service Created
Book Repository : Fetching books..
After executing : displayService
Book Repository : Fetching books..

Process finished with exit code 0
```

EXERCISE 4:



```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    https://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion> Compatible with Maven 3

  <groupId>com.library</groupId>
  <artifactId>LibraryManagement</artifactId>
  <version>1.0</version>

  <properties>
    <maven.compiler.source>1.8</maven.compiler.source>
    <maven.compiler.target>1.8</maven.compiler.target>
  </properties>

  <dependencies>

    <!-- Spring Context -->
    <dependency>
      <groupId>org.springframework</groupId>
      <artifactId>spring-context</artifactId>
      <version>5.3.39</version>
    </dependency>

    <!-- Spring AOP -->
    <dependency>
      <groupId>org.springframework</groupId>
```

```

<build>
  <plugins>
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-compiler-plugin</artifactId>
      <version>3.11.0</version>
      <configuration>
        <source>1.8</source>
        <target>1.8</target>
      </configuration>
    </plugin>
  </plugins>
</build>
</project>

```

EXERCISE 5:

```

1 package org.project.librarymanagementapplication;
2
3 import org.project.librarymanagementapplication.repository.BookRepository;
4 import org.project.librarymanagementapplication.service.BookService;
5 import org.springframework.boot.SpringApplication;
6 import org.springframework.boot.autoconfigure.SpringBootApplication;
7 import org.springframework.context.ApplicationContext;
8 import org.springframework.context.support.ClassPathXmlApplicationContext;
9
10 import org.springframework.boot.autoconfigure.SpringBootApplication;
11 import org.springframework.boot.SpringApplication;
12
13 @SpringBootApplication
14 public class LibraryManagementApplication {
15
16     public static void main(String[] args) {
17         ApplicationContext context= new ClassPathXmlApplicationContext("applicationcont
18
19         BookService service =context.getBean( name: "bookService",BookService.class);
20
21         BookRepository repo=context.getBean( name: "bookRepository", BookRepository.class);
22
23         service.showBooks();
24
25
26
27         //SpringApplication.run(LibraryManagementApplication.class, args);
28     }
29
30 }
31

```

```
Before executing : showBooks
Book Service Executed
Fetching book Details from Repository...
After executing : showBooks

Process finished with exit code 0
```

```
package org.project.librarymanagementapplication.repository;

public class BookRepository { 7 usages
    public void getBookDetails(){ 1 usage
        System.out.println("Fetching book Details from Repository...");
    }
}
```

```
package org.project.librarymanagementapplication.service;

import org.project.librarymanagementapplication.repository.BookRepository;

import java.awt.print.Book;

public class BookService { 6 usages

    private BookRepository bookRepository; 2 usages
    public void setBookRepository(BookRepository bookRepository){
        this.bookRepository=bookRepository;
    }
    public void showBooks(){ 1 usage
        System.out.println("Book Service Executed");
        bookRepository.getBookDetails();
    }
}
```


EXERCISE 6:

```
package org.project.librarymanagementapplication;

import org.project.librarymanagementapplication.service.BookService;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class LibraryManagementApplication {

    public static void main(String[] args) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");

        BookService service =
            context.getBean("bookService", BookService.class);

        service.showBooks();
    }
}
```

```
Book Service Executed
Fetching Book Details from Repository...

Process finished with exit code 0
```

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
           http://www.springframework.org/schema/beans
           https://www.springframework.org/schema/beans/spring-beans.xsd">

    <!-- Repository Bean -->
    <bean id="bookRepository"
          class="org.project.librarymanagementapplication.repository.BookRepository"/>

    <!-- Service Bean -->
    <bean id="bookService"
          class="org.project.librarymanagementapplication.service.BookService">

        <!-- Setter Injection -->
        <property name="bookRepository"
                  ref="bookRepository"/>

    </bean>

</beans>
```

```

package org.project.librarymanagementapplication.service;

import org.project.librarymanagementapplication.repository.BookRepository;
import org.project.librarymanagementapplication.service.BookService;

public class BookService { 5 usages

    private BookRepository bookRepository; 2 usages

    public void setBookRepository(BookRepository bookRepository) { no usages
        this.bookRepository = bookRepository;
    }

    public void showBooks() { 1 usage
        System.out.println("Book Service Executed");
        bookRepository.getBookDetails();
    }
}

```

EXERCISE 7:

```

Constructor Injection Executed
Setter Injection Executed
Book Service Started
Fetching Book Details from Repository..
Process finished with exit code 0

```

```

package org.project.librarymanagementapplication;

import org.project.librarymanagementapplication.service.BookService;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class LibraryManagementApplication {

    public static void main(String[] args) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");

        BookService service =
            context.getBean("bookService", BookService.class);

        service.showBooks();
    }
}

```

```

<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
         http://www.springframework.org/schema/beans
         https://www.springframework.org/schema/beans/spring-beans.xsd">

    <!-- Repository Bean -->
    <bean id="bookRepository"
        class="org.project.librarymanagementapplication.repository.BookRepository"/>

    <!-- Service Bean -->
    <bean id="bookService"
        class="org.project.librarymanagementapplication.service.BookService">

        <!-- Constructor Injection -->
        <constructor-arg ref="bookRepository"/>

        <!-- Setter Injection -->
        <property name="bookRepository"
            ref="bookRepository"/>

    </bean>
</beans>

```

```

package org.project.librarymanagementapplication.service;

import org.project.librarymanagementapplication.repository.BookRepository;

public class BookService { 6 usages

    private BookRepository bookRepository; 3 usages

    public BookService(BookRepository bookRepository) { no usages
        this.bookRepository = bookRepository;
        System.out.println("Constructor Injection Executed");
    }

    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
        System.out.println("Setter Injection Executed");
    }

    public void showBooks() { 1 usage
        System.out.println("Book Service Started");
        bookRepository.getBookDetails();
    }
}

```

EXERCISE 8:

```
Before Method : showBooks  
Book Service Started  
Fetching Book Details...  
After Method : showBooks  
  
Process finished with exit code 0
```

```
package org.project.librarymanagementapplication;  
  
import org.project.librarymanagementapplication.service.BookService;  
  
import org.springframework.context.ApplicationContext;  
import org.springframework.context.support.ClassPathXmlApplicationContext;  
  
public class LibraryManagementApplication {  
    public static void main(String[] args) {  
        ApplicationContext context =  
            new ClassPathXmlApplicationContext("applicationContext.xml");  
  
        BookService service =  
            context.getBean("bookService", BookService.class);  
  
        service.showBooks();  
    }  
}
```

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xmlns:aop="http://www.springframework.org/schema/aop"
       xsi:schemaLocation="
           http://www.springframework.org/schema/beans
           https://www.springframework.org/schema/beans/spring-beans.xsd
           http://www.springframework.org/schema/aop
           https://www.springframework.org/schema/aop/spring-aop.xsd">

    <!-- Repository Bean -->
    <bean id="bookRepository"
        class="org.project.librarymanagementapplication.repository.BookRepository"/>

    <!-- Service Bean -->
    <bean id="bookService"
        class="org.project.librarymanagementapplication.service.BookService">

        <property name="bookRepository"
            ref="bookRepository"/>

    </bean>

    <!-- Aspect Bean -->
    <bean id="loggingAspect"
        class="com.library.aspect.LoggingAspect"/>

    <!-- Enable Spring AOP -->
    <aop:aspectj-autoproxy/>

</beans>
```

EXERCISE 9:

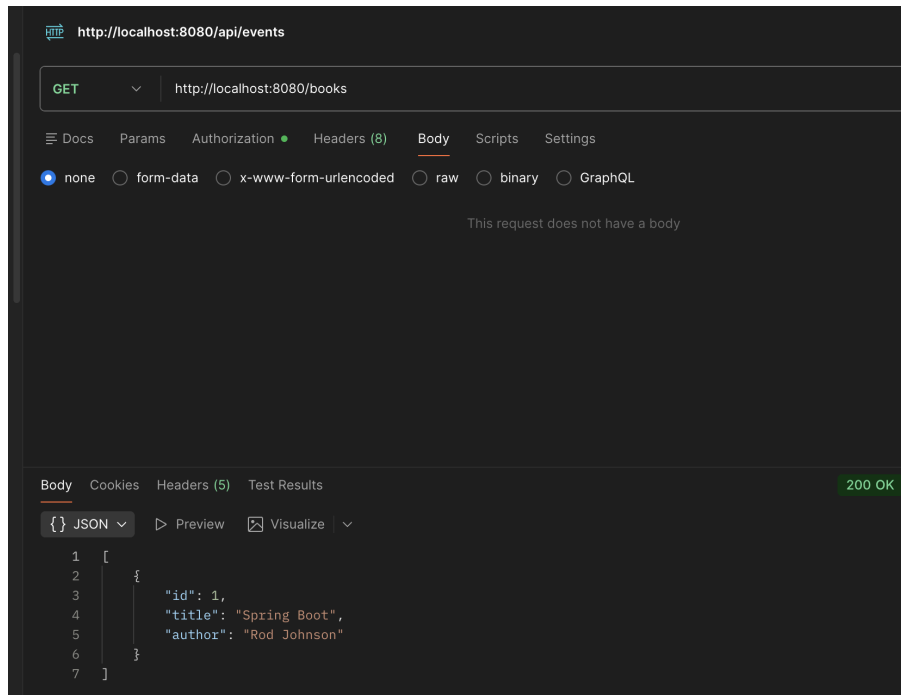
The screenshot shows a REST client interface. At the top, a dropdown menu is set to 'POST' and the URL is 'http://localhost:8080/books'. Below this, there are tabs for 'Docs', 'Params', 'Authorization', 'Headers (10)', 'Body', and 'Scripts'. The 'Body' tab is selected, and the body type is set to 'raw'. The body content is a JSON object:

```
1 {  
2   "id":1,  
3   "title":"Spring Boot",  
4   "author":"Rod Johnson"  
5 }
```

Below the main interface, there is a section for 'Body', 'Cookies', 'Headers (5)', and 'Test Results'. The 'Body' tab is selected, and the body type is set to 'JSON'. The body content is the same JSON object:

```
1 {  
2   "id": 1,  
3   "title": "Spring Boot",  
4   "author": "Rod Johnson"  
5 }
```

```
package org.project.librarymanagementapplication;  
  
import org.springframework.boot.SpringApplication;  
import org.springframework.boot.autoconfigure.SpringBootApplication;  
  
@SpringBootApplication  
public class LibraryManagementApplication {  
  
    public static void main(String[] args) {  
  
        SpringApplication.run(  
            LibraryManagementApplication.class,args);  
  
    }  
  
}
```



```
package org.project.librarymanagementapplication.repository;

import org.project.librarymanagementapplication.entity.Book;
import org.springframework.data.jpa.repository.JpaRepository;

public interface BookRepository 4 usages
    extends JpaRepository<Book,Integer> {
}
```

```

package org.project.librarymanagementapplication.entity;
import jakarta.persistence.Entity;
import jakarta.persistence.Id;

@Entity
public class Book {
    @Id
    private int id;
    private String title; 3 usages
    private String author; 3 usages
    public Book() {
    }

    public Book(int id, String title, String author) { no usages
        this.id=id;
        this.title=title;
        this.author=author;
    }

    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id=id;
    }

    public String getTitle() { no usages
        return title;
    }

    public void setTitle(String title) { no usages
        this.title=title;
    }

    public String getAuthor() { no usages
        return author;
    }

    public void setAuthor(String author) { no usages
        this.author=author;
    }
}

```

```

package org.project.librarymanagementapplication.controller;
import org.project.librarymanagementapplication.entity.Book;
import org.project.librarymanagementapplication.repository.BookRepository;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.*;

import java.util.List;

@RestController
public class BookController {

    @Autowired
    private BookRepository repository;

    @PostMapping("/books")
    public Book addBook(@RequestBody Book book){
        return repository.save(book);
    }

    @GetMapping("/books")
    public List<Book> getBooks(){
        return repository.findAll();
    }
}

```